JOURNAL OF EXPERIMENTAL SOCIAL PSYCHOLOGY 1, 20–54 (1965)

Collective Behavior in a Simulated Panic Situation[1]

HAROLD H. KELLEY, JOHN C. CONDRY, JR.
University of California, Los Angeles

ARNOLD E. DAHLKE
University of Oklahoma

AND

ARTHUR H. HILL
University of Texas

The experiments reported here are concerned with the behavior exhibited by a collection of persons when they are suddenly confronted with an imminent danger and provided with a means of escape that is limited and for which they do not possess any prior organized mode of use. Because the escape avenue permits only one person at a time to escape, they are contriently interdependent (Deutsch, 1949). One person's use of the escape route reduces the likelihood that the remaining individuals will be able to escape. Furthermore, simultaneous attempts to use the route reduce its effectiveness. Under these conditions, the escape efforts of the individuals may be orderly, permitting the maximum number possible for the available time to traverse the escape route. Or, the escape efforts may be disorderly and characterized by "traffic jams" which waste time and reduce the number who succeed in escaping. We are concerned here with some of the factors which affect the extent to which the collection's behavior is incoordinated. It can be seen that we are examining a collective situation very similar to that studied by Mintz in his widely known experiment (1951) which Brown (1954), in his excellent review of the literature on mass phenomena, has described as a laboratory paradigm of an escape mob.

Although not in complete agreement, writers on panic before Mintz had tended to emphasize *perceived danger* and *mutual influence* (suggestion, contagion, mimicry) as the key factors in the development and spread of incoordinated and nonadaptive "panic" behavior. In the emotional state induced by the threat of the impending danger, indi-

[1] The research reported here was made possible by two National Science Foundation grants to the senior author: G-5553 (University of Minnesota) and G-18468 (University of California, Los Angeles).

viduals were thought to become more suggestible and, hence, more influenced by the example of others' behavior and less governed by ordinary critical thought processes, moral codes, etc. Taking issue with this view, Mintz emphasized that persons are merely reacting to the reward structure of the situation. As long as escape is orderly, each sees that his interests lie in continuing to cooperate with others. However, when a few individuals begin to create a disturbance in the use of the exit, the others will recognize the threat to their own escape and see that their only hope lies in themselves trying to push through. Thus, the individual is no less rational or moral in the panic than in any other situation. He is always in pursuit of his own interests and acts on the basis of his current estimates of where these lie. In this analysis, Mintz is most sharply in conflict with the earlier views on the question of the role of perceived danger. Mintz explicitly denies that intense fear (or emotional excitement, as he often refers to it) is responsible for nonadaptive group behavior.

While the originality of Mintz's work must be acknowledged, several major criticisms can be made of his procedure and conceptualization. The most important of these is that his method cannot be considered to test the effects of danger. The subjects were promised small monetary rewards, ranging from 10¢ to 25¢, for escaping (withdrawing their cones from within a large bottle) in time, and even smaller fines, from 1¢ to 10¢, for failing to do so. It is difficult to believe that the results from this situation shed much light on behavior under conditions of high stress. To argue, as Mintz does, that intense fear is shown not to be important because even in its absence there occurs "behavior analogous to that occurring in panics" is to overlook the possibility that collective incoordination may have a variety of quite different causes.

A second criticism, first noted by Brown (1954), is that Mintz's analysis of incoordination is only a partial one. It starts at a point in time when a disturbance or jam has already begun. It does not face the problem of the *origination* of the first jam-producing escape attempts.

A third criticism is that the behaviors of Mintz's subjects do not seem to provide apt illustrations of his view that even under the press of the "panic," individuals continue to act in a thoughtful manner. This view certainly provides no explanation of why a subject should, on seeing two other cones jammed in the bottleneck, nevertheless immediately proceed to try to remove his own cone. This is particularly difficult to understand in view of the fact that he had just been given a convincing demonstration that only one cone can be removed at a time.

Finally, if we look to Mintz's study for practical recommendations about how to reduce incoordination in danger situations (and practical implications are surely one goal of science), none is forthcoming. Among his several experimental variations, the only one in which the amount of incoordination was markedly reduced involved group goals (surpassing the performance of a group from another university) rather than individual ones (gaining individual rewards and avoiding individual fines). Limited to this result one would be forced to conclude that incoordination is more or less inevitable when members of a collectivity are oriented toward their separate personal goals, and that improvement is possible only when these orientations are supplanted by the identification with success of the total group. This provides a totally misleading emphasis. Surely there must be some factors within the individual reward-fine situation that have important effects upon the amount of jamming. The present criticism is that Mintz's work fails to reveal these and, worse, implies that none of any significance exists.

The present experiments were designed to pursue the question of the factors and processes underlying incoordination in the interdependent escape situation. As will be seen, the theoretical analysis and experimental procedures were planned to meet some of the above objections to Mintz's work. The analysis deals with the origin as well as the spread of escape attempts, the latter being cast in terms of a more modern theory of social influence. The experimental procedures deal exclusively with the case where "selfish" individual interests predominate and focus upon some of the factors within that setting which affect the collectivity's success. Most important among these is degree of threat or danger. Levels of threat as high as seemed experimentally feasible were examined and compared with the absence of threat.

We shall first consider the general conceptual framework, and then the methods and results of the three experiments in turn.

THEORETICAL ANALYSIS

It would seem that a complete account of collective performance in the interdependent escape situation would include (1) a conception of what constitutes a "solution"[2] to the problem posed for the collectivity and (2) a theory of individual behavior in the escape situation. The first would specify an ideal pattern for the various participants' behavior, and the second would suggest the factors determining the extent to which this pattern would be realized.

The nature of the solution in this case is easy to specify. It is what

[2] "Solution" is used here in the same sense as in the outline of the 1952 edition of the *Readings in Social Psychology* by Swanson, Newcomb, and Hartley (1952). The Mintz study was placed in part II dealing with collective problem-solving and in a subsection entitled "Behavior in the absence of collective solutions."

the participants would try to arrange if they had full knowledge of the problem and time to analyze, discuss, and agree upon a method for coping with it. They would plan, obviously, a queue. The first person in the queue would use the escape route first while the others waited, the second would use it next, and so on down the line. The concept of queue affords a criterion against which to evaluate the performance of the collectivity. This criterion is used implicitly when "time to escape" or "jamming" are employed as proficiency measures. Both increase with increasing departures from an orderly queueing procedure.

To say that the participants might *try* to form and maintain a queue is not, of course, to say they would succeed in doing so. It is instructive to consider why their plan might break down. The answer is that certain persons in the queue would not be satisfied with the positions they were allotted and would try to occupy positions closer to the head of the line. In doing so, they would interfere with the escape of persons with higher priorities and consequently waste time for the queue.

This suggests something about the conditions under which a queue might spontaneously develop and persist, even, as in the case under study, in the absence of prior opportunities for planning. A queue would be likely to develop if there were great variability among the individuals in their attitudes toward the situation—in how urgent they felt the necessity of escape to be. Given heterogeneity of attitudes, the person feeling the greatest urgency would go first while the others more or less willingly waited, followed by the person with the next greatest concern, and so on, with the individual most indifferent to the threatened danger being willing to wait until the end of the line. An orderly queue would also be likely to develop if all participants held attitudes of indifference. In this case, some time might be lost while all waited for someone else to go first, but assuming a general weak incentive for leaving the situation (as in a practice fire drill), this type of time wasting would probably be of little importance. They certainly would not waste time with jams.

The distribution of attitudes most obviously detrimental to the development and maintenance of a queue is that of uniformly high felt urgency. If all judge the danger to be imminent and intolerable, and are, consequently, highly desirous of immediate escape, all will seek the head of the queue and none will be willing to accept positions toward the rear.

This analysis assumes that a person's behavior with respect to attempting immediate escape vs. waiting for others to escape depends upon his assessment of the situation, i.e., his attitude toward it. Accordingly, the key factor mediating the success or failure of a collectivity is the *distribution* of these attitudes. If they are concentrated at the high urgency end of the scale, either being so from the outset or becoming so during the escape period, the occurrence of orderly queueing is unlikely. We must consider, therefore, factors that will bias the initial distribution toward the anxious extreme or that will make for shifts in that direction.

The latter eventuality, that the distribution of attitudes may shift toward the high urgency end of the scale, implies changes in individuals' evaluations of the situation as a result of events in the early stages of the escape period. Let us briefly consider how this might occur. One possibility is that the person's estimate of the likelihood of successful escape may change as time passes, and particularly as time is wasted by jamming. In this case, the actions of the others *indirectly* affect his attitudes through changing the characteristics of the situation in several crucial respects, viz., the time left and the apparent inefficiency of time use.

A second possibility, and one much more familiar to the social psychologist, is that the actions of others *directly* affect his attitudes. Much of what is known from research on interpersonal influence suggests that the conditions in the typical interdependent escape situation are highly appropriate for such influence to occur. The situation is an *ambiguous* one, difficult for the individual to assess. The information available is unclear with regard to how dangerous the threatened consequences are and their imminence. The situation is also an *important* one. He must make some sort of judgment and decide upon his action regarding the escape, and the wisdom of this action may have important consequences for him. Finally, the situation permits *communication* among the participants. Information about others' evaluations is communicated to the individual through their verbal behavior and escape actions, or in our experiments where verbal communication is ruled out, through their behavioral examples alone. Given these conditions of ambiguity, importance, and communication, individuals will be influenced by one another and this influence process will operate to produce a uniformity or concentration of attitudes. The entire process can be conceptualized in terms of a group powerfield that produces similarity among the participants in their life spaces, as suggested by French (1944), or in terms of pressures toward uniformity, as described by Festinger (1950).

In brief, we are arguing that the distribution of attitudes toward the escape situation is the critical factor in the performance of the collectivity. If this distribution is initially, or later becomes, concentrated at the end of the attitude continuum reflecting high concern and feelings of urgency about escape, the spontaneous formation and maintenance of an orderly queue will not be possible. Each variable in the escape situation may then be examined for (1) its possible effect upon the initial distribution of attitudes, and (2) its possible contribution to changes in these attitudes during the course of the escape process.

In the experiments to be described, we have studied four experimental variables. The first is the *threatened penalty* for failure to escape. With increasing magnitude of penalty, increased concentration in the state of high concern can be expected for several different reasons. First, the most direct and obvious effect is that the greater the threat, the greater the number of participants who will be highly concerned (and realistically so) and desire to be first in the escape queue. Second, as a byproduct of the first effect, the higher threat will produce more time-wasting jams which will demonstrate to initially unconcerned persons that successful escape through waiting is less promising than they might have thought. Third, as an additional byproduct of the first, the greater number of escape attempts generated by high threat will constitute a greater volume of communication conveying the evaluation of the situation as one requiring urgent action. Fourth, in addition to being more numerous, these communications may also exert more influence upon the other participants because of the heightened importance of the situation, following Festinger's (1950) hypothesis that pressures toward uniformity increase as the issue becomes more important. The last three points all refer to ways in which high threat may increase the likelihood that initially less concerned persons will be swayed toward greater concern by the behavior of those attempting escape. For all these various reasons, the threat of large penalties for failure to escape should produce more incoordination than the threat of small penalties.

The second factor is *size* of the collectivity. The effect of size is likely to be, in part, purely mathematical: the greater the number of persons involved, the greater the likelihood that two or more will feel a similar degree of high urgency, attempt to escape at the same time, and prevent each other from doing so. In addition, size may have a preinteraction psychological effect: the more people involved, the more likely are members to judge the available time (whatever it is estimated to be) to be insufficient and, hence, early escape to be important. Finally, as outgrowths of the above effects, large group size may be expected to produce second-order effects during the interaction which are similar to those suggested above for high threatened penalty. The high rate of escape attempts and jams will act to change the attitudes of other participants, through providing new information about the situation (regarding the time available and efficiency of time use) and through communicating evaluations of the situation that may influence the initially unconcerned or undecided. All these considerations indicate that large collectivities should show greater incoordination in their escape efforts than small ones.

Our third and fourth experimental variables were formulated explicitly in terms of the social influence process occurring during interdependent escape. The third variable is the individual's *susceptibility to social influence:* whether he is "set" to take his cues from the other participants and be influenced by them, or to be independent and make up his own mind about the situation. The former should increase the concentration or homogenization of views and the latter should retard it. The effect of this would be expected to depend upon what initial attitude is prevalent within the collectivity. If the distribution is initially weighted toward the high concern end, being oriented toward basing one's evaluation upon others' behavior should make for numerous and rapid shifts toward that end, with highly detrimental consequences for group coordination.

Our fourth and final experimental variable is the *availability of confidence responses.* This might be viewed in terms of anti-panic leadership (Brown, 1954). It is often true in danger situations that the highly confident and calm people have no way to express their evaluation of the situation in a manner as dramatic and unequivocal as are the behaviors (i.e., the escape attempts) of the highly concerned persons. Under these conditions, the actions of the latter are likely to catch the attention of the less worried or uncertain participants, and, in the absence of counterbalancing examples from the opposite extreme, sway them in the direction of high concern. The analogy might be drawn to a group discussion where the proponents of only one extreme viewpoint can find words to express their position. The tendency would be for the opinion distribution to shift toward the expressed position. In our basic experimental situation, each person has only two response options: to wait or to attempt to escape. Therefore, the most confident persons are not distinguishable from other "waiters" who may be quite worried but still willing, at least momentarily, to wait. In a special three-response condition, an additional action was provided whereby individuals could indicate their confidence and willingness to let others go first. On the assumption that use of this response would prevent concentration at the anxious end of the attitude continuum, it was predicted that collective coordination would be greater than in the two-response condition.

GENERAL PROCEDURE

In the three experiments to be reported, the situation and procedure were essentially the same from one to the next. In all cases, the *Ss* came to a laboratory at a scheduled time and were seated in separate booths or rooms. They were not able to see and were not permitted to speak to one another during the experiment. Upon being seated, each one had electrodes, for delivering shock, attached to the first and third fingers of his non-preferred hand.

The experiment was described as a study of behavior under threat, the situation being one where a number of people have to use a single, limited exit to escape from an impending danger within a limited time. The threatened penalty for failure to escape was, in most cases, one or more painful electric shocks. The fact that only one person at a time could escape was made clear by explanation and demonstration.

While the details differed from one experiment to the next, in all cases the *S* could see signal lights showing his own position vis-à-vis the danger situation and similar lights showing the position of each of the other *S*s. At the beginning of a trial, each one's light indicated he was in "danger." By making a simple response (turning or pressing a switch), a person could attempt to escape. When he did so, he changed the color of the signal lights corresponding to him, and in this manner his escape attempt was made visible to all the others (and himself, as well). When one *S*, and only one, has his escape switch "on" for a brief time (3 seconds in all cases), his signal lights turned green, which indicated for all to see that he had succeeded in escaping from the danger. Interdependence in escape was introduced by the fact that if two or more *S*s had their escape switches "on" at the same time, none could escape and they all remained in danger. In other words, one person had to have sole and uninterrupted occupancy of the escape "route" for a 3-second period if he was to escape. Successful escapes were shown to the *S*s (with illustrations of the 3-second period but without labeling its duration) as were mutual interferences or "jams."

Following these instructions, the experimental trials were run, *E* giving "start" and "stop" signals to provide a standard time period for escape. After the trials, the *S*s were given posttrial questionnaires followed by a "catharsis" session in which the experiment was explained, justification was given for subjecting them to threat, and they were permitted to ask questions and air their feelings. Finally, they were asked not to discuss the experiment with other potential *S*s, a request to which they all agreed and with which they apparently complied.

Although it was designed to create the same type of "bottleneck" situation, there are essential differences between the apparatus used in these experiments and that used by Mintz (1951). First, the *S*s in these experiments were isolated from one another, there being little possibility for visual or verbal contact during the trial. This provided better control of the interpersonal variables that might operate in such a situation. Since an *S* could not tell the identity of persons attempting to escape, such variables as personality impressions and clues as to social status could not affect his reactions to these events.

A second essential difference is in the use of electronic equipment. In Mintz's study it was possible for the cones to jam tightly in the bottleneck and delay the trial while they were being dislodged. This was not possible in this experiment. The escape and interdependence were controlled by electronic circuits which enabled a jam to be disentangled merely by the participants' releasing their escape switches. Thus, the time loss due to jams was entirely under the control of the participants. Further, it was possible to obtain precise data on the process of escape. When any *S* pressed his escape switch, this action was automatically recorded on an Esterline-Angus operations recorder. This enabled the *E* to return to the data and ascertain such things as how many jams occurred and for what duration, which members were involved in each jam, the exact pattern of events leading to jams, etc.

The latter aspect of our apparatus also enabled us to use a different measure of group incoordination than Mintz had employed. He reports only whether or not a jam developed within each of the collections in his various experimental conditions. We used as the basic dependent variable the percentage of persons succeeding in escaping during a standard time period. This particular measure, although easily determined and interpreted, seems rather crude at first glance. It was chosen when two other more refined measures were found to be highly correlated with it. The first such measure, termed "man-seconds," was the sum over the various *S*s of the time each one spent in the danger situation until his escape or, if he failed to escape, until the end of the trial. It reflects (inversely) not only the number of persons who succeeded in escaping, but how soon in the trial they did so. A second measure, termed "jam-seconds," was the sum over the various *S*s of the time each one had his escape switch "on" while one or more others also had theirs "on." In other words, it reflects the length of jams and the number of persons contributing to them. These two indices and the percentage of persons escaping during the first trial were derived from the data obtained in Experiment III. Linear relationships were found among the three variables, and the percentage of persons escaping yielded product moment correlations of -0.97 and -0.92, respectively with "man-seconds" and "jam-seconds" ($N = 40$ in each case). This was taken as sufficient reason for using the simple percentage of escapes as our sole measure of group coordination.

It may further be noted that in two of the experiments (I and III), the *S*s were put through three successive trials. Throughout this paper we rely exclusively upon data from the first trial. This decision is based upon recognition of the difficulties involved in interpreting the behavior on the later trials created by such probable factors as incredulity (when a promised threat is not delivered), practice effects, and physiological changes (when drugs are used as in Experiment I).

EXPERIMENT I[3]

Procedure

Design and subjects. The first experiment to be reported[4] dealt with two of the factors discussed earlier: (1) magnitude of the threatened penalty and (2) size of the collectivity. The three magnitudes of penalty used will be referred to as *low threat, medium threat,* and *high threat.* Collections of *four, five, six,* and *seven* *S*s were employed. Each collection was composed of *S*s of the same sex, either all males or all females. Thus, the design was a $3 \times 4 \times 2$ factorial design with two collections assigned to each condition. It can be seen that the total number of collections was 48 and the total number of *S*s, 264. The collection is the unit upon which statistical analysis is based.

Collections in the low- and medium-threat conditions were studied during one quarter of the academic year, and those in the high-threat condition, during the following quarter. All *S*s were drawn from the same supply source, namely, introductory psychology students at the University of Minnesota who had indicated a willingness to participate in psychological experiments. Because they receive two extra points on their final examination grade for each hour of experimental participation, about 90% of the students enrolled in the introductory classes express willingness to serve as experimental *S*s and are placed in a common subject pool. For the low- and medium-threat conditions, *S*s were selected from this pool simply on the basis of their conformity to sex and scheduling requirements. They were notified by postcard of the time and place of the experiment, but no inkling was given of its nature. Insofar as possible, the various conditions were run in random order, but exceptions were made when fewer *S*s appeared than had been scheduled.

The selection of *S*s for the high-threat condition was based upon an additional criterion. In order to insure that no harmful effects would result from the injection of epinephrine, the records of *S*s meeting the sex and schedule requirements were cleared with the University's Student Health Service. Only those persons judged to be able to withstand the injection without any possibility of ill effects (and this was the vast majority) were contacted for the experiment.

Apparatus. The *S*s were seated in booths which enabled them to see the *E* at the front of the room but not one another. Mounted in front of each was an escape

[3] This experiment was conducted by Dahlke in the Laboratory for Research in Social Relations, University of Minnesota.

[4] The three experiments are presented in logical order rather than in the order in which they were performed. The third one reported here was actually the first in the series. It was followed, in order, by the ones described here as Experiments I and II.

switch which they were to move when they wished to escape. A large panel of lights, placed in the front of the room so that all *S*s could see it, displayed the actions of all the participants. By watching the light panel, each *S* could determine what every other *S* was doing at any point in time, whether each had succeeded in escaping, etc.

The light panel displayed a column of three colored lights for each *S*. At the start of the experiment, a white light was "on" in each column indicating that each person was in "danger." When an *S* moved his escape switch to the right, a red light appeared in his column on the panel indicating an attempted escape. If the escape was successful, a green light appeared in his column. The columns were labeled A through D (or up to G, if necessary) but in mixed order. Each *S* knew his own letter and therefore could readily locate the column of lights displaying his actions and fate. While he could see the three to six columns corresponding to the other *S*s, because of the mixed order of the labels, he could not determine which of them corresponded to any particular other booth or *S*. Thus, even though he may have formed an impression of another *S* and noted his booth location, the individual could not tell which lights indicated the other's actions.

The time necessary for a single *S* to escape was 3 seconds, provided no other *S*s attempted to escape during this time. If two or more attempted to escape at the same time, all remained in the "red"condition and none escaped until all but one person had released their escape switches, and he had sole possession of the "red" (the "escape route") for 3 continuous seconds. The 3-second and jam rules were applied by a research assistant located in an adjoining control room.[5] Guided by a metronome beat at half-second intervals and possessing a light panel similar to that visible to the *S*s, the assistant turned on a given *S*'s green light when that *S* alone had his escape light on for 3 uninterrupted seconds. When the green light showing successful escape was turned on, that *S*'s switch was deactivated so that his subsequent actions had no effect on the lights or upon others' escape attempts.

The time allowed for the entire collection to escape was determined by the flow of red-colored water from one large water bottle into another, in a water version of an hour-glass type timer. This was done so the *S*s would have no reliable index of the length of time available but at the same time would have a vivid reminder that time was "running out." The *S*s were told:

> "The time you have in which to escape is the time required for the water to flow from the top bottle of these two to the bottom bottle. I merely reverse their positions. When all of the water has left the top bottle, I will announce that time is up."

The amount of water in the bottles was varied to provide twice as much time as was minimally necessary for the given number of *S*s to escape. Thus, with the minimum escape time of 3 seconds per person, a maximally efficient four-man group would require 12 seconds, and was therefore allowed 24 seconds. With the same rule, five-man groups were allowed 30 seconds; six-man groups, 36 seconds; and seven-man groups, 42 seconds. The *S*s were not told how long they would have, their only clue to this being the volume of water present in the bottles at the start.

Instructions. When the *S*s entered the experimental room, they were seated and instructed not to talk. In the medium- and high-threat conditions, electrodes were taped to the first and third fingers of each one's non-preferred hand, and the *E* determined each one's "shock threshold." He also delivered two fairly strong shocks

[5] We are indebted to Lee A. Borah, Jr., for his careful performance of this task

(one of approximately 40 volts, 60-cycle ac, and the other of about 50 volts) and noted the *S*'s reactions to them. These "data" were referred to later when the *S*s were threatened with more severe shocks. Following this, *S*s in the high-threat condition were given a subcutaneous injection of epinephrine by a physician[6] while *E* gave the following explanation:

> "The shot you are getting is a neutral solution. Last quarter we were interested in what effect vitamin B would have in this experimental situation, so we injected all our subjects with vitamin B. When we analyzed our results we couldn't determine whether they were due to the vitamin or just to getting the shot itself. So, we are now using you as 'control' subjects and are giving you a completely neutral solution. This will have no physiological effects—it is completely neutral."

The *S*s in the low-threat condition were not attached to electrodes and received neither shocks nor injections.

After the above preliminaries, the basic instructions were given. The experiment was described as part of an investigation supported by a federal agency for the purpose of studying the "alarming increase in the rate of deaths and injuries occurring in civilian disasters" and of solving the problem of the "causative factors" in these situations. The *S*s were told:

> "The results we obtain will make a very important contribution to its solution. I'd like to ask all of you for your utmost cooperation. You people are going to be placed in a (danger) situation and given a limited amount of time to escape. However, it takes a short, but definite time interval for any one individual to escape. If another person tries to escape at the same time, he will become, so to speak, jammed in the exit, and neither will be able to escape. Such jams will waste some of the time available for escape." (The word "danger" was omitted from the instructions given the low-threat *S*s.)

The *S*s were then shown how to use the apparatus, and demonstrations were given of an escape, jam, and resolution of a jam. Guided by *E*'s directions, two persons tried to escape at the same time and a jam was demonstrated. Then, one was directed to release his switch and the other's escape was shown. Then the former was permitted to escape. The time allowed for the entire group to escape was explained as above.

In addition, the medium- and high-threat *S*s were told:

> "I have called this a danger situation because those of you who do not escape before the time is up will receive a painful electric shock as a result of not escaping. Just a few moments ago, we determined a "shock threshold" for each of you. We went through all the trouble of getting these measurements because we want to be able to give you the most painful and uncomfortable shock we can, without seriously hurting you in any way. I want to be completely honest with you and tell you exactly what you are in for. Those of you who do not escape will receive a series of extremely painful shocks. They'll be a lot more painful

[6] The dosage and procedure here were identical with those used by Schachter and Singer (1962). The dosage was ½ cc of a 1:1000 solution of Winthrop Laboratory's Suprarenin. We are indebted to Dr. Abelardo Mena, M.D., for giving the injections and providing the necessary safeguards in the event any *S* had overreacted to the injection or the drug.

than the ones we gave you a few moments ago—they're going to hurt. But we must do this to impress upon you the uncomfortable consequences of not escaping. By combining the value of your shock threshold with your ability to discriminate between two shocks at a higher level, we have been able to estimate how much shock would be dangerously harmful to you. We're not going to give you that amount, of course, but we are going to get as close to it as we possibly can. Those of you who do not escape will have to face the consequences of as painful and uncomfortable a shock as we can give."

Finally, all groups were told:

"About this time in past sessions of this experiment some people expressed a desire to leave, after finding out what it was all about. But I'm afraid I can't allow this. . . . However, I would like you to indicate on these questionnaires how you feel about being in the experiment and whether or not you feel like leaving at this time."

The pretrial questionnaire was given to *S*s in all conditions. This required them to state whether they felt like leaving the experiment, to rate their feelings about being in the experiment on a six-point scale from "extremely calm" to "extremely uneasy," and to estimate how many of their group would succeed in escaping. Immediately following these questions, the first trial was run. In the medium- and high-threat conditions, the *S*s who did not escape this time were told they would be shocked after the last trial in the series, and a second trial was run. The same procedure was repeated for a third trial after which the *S*s were given a final questionnaire. Needless to say, they were never given the threatened shocks. The final phase of the procedure was a "catharsis" session and request for secrecy.

The reader will note particularly the difference between the three threat conditions. The *low-threat* *S*s faced no penalty for failing to escape. The instructions simply indicated they were to try to escape during the limited time period. The *medium-threat* *S*s were shown convincingly that they could be given electric shocks, and they were threatened with the penalty of as painful shocks as possible (without harming them) if they failed to escape. The *high-threat* *S*s were in the same situation except that they had heard the threat of painful shocks from 7 to 10 minutes after having received the injection of epinephrine. This was assumed to make for a higher level of anxiety on the theory, as stated by Schachter and Singer (1962), that the strength of an emotional reaction is a joint function of the external stimulus conditions (the same threat for out medium- and high-threat *S*s) and the degree of excitation of the sympathetic nervous system (higher for the high-threat *S*s because of the injection). Schachter's pretests had shown that the sympathetic effects of the given dosage of epinephrine begin within 3–5 minutes after the injection and last for at least 10 minutes. The timing of our procedure made it highly likely that all *S*s were experiencing symptoms often associated with danger (speeded heart rate and involuntary tremor) during the time they were being told of the painful consequences of failure to escape and during the escape trials.

Results

As a check upon the efficacy of the threat manipulations, the *S*s in all conditions were asked, in the pretrial questionnaire, if they wished to leave the experiment. The results are presented in Table 1. Because the data did not vary with size, the numbers expressing a wish to leave are shown only for the males and females within each threat condition. It can be seen that considerably fewer males indicated a desire to leave, and in general, the results indicate a difference between the threat conditions. The high-threat groups do not differ greatly from the medium-threat groups, but there is a sizable difference between low threat and the other two conditions.

TABLE 1
NUMBER OF SUBJECTS EXPRESSING DESIRE TO LEAVE EXPERIMENT BEFORE FIRST TRIAL (EXPERIMENT I)

	Threat condition		
Sex	Low	Medium	High
Males	0	3	4
Females	2	12	11

There are forty-four *S*s in each of the six cells in Table 1. It can be seen that we succeeded in creating enough concern on the part of the females in the medium and high conditions that one in four was willing to sacrifice whatever time and energy investment had been made in coming to the experiment in order to withdraw from it.

A further check upon the threat manipulations is provided by the pre-trial ratings of uneasiness. As in the case of the data above, these ratings did not vary systematically with size so the averages are presented in Table 2 only by sex and threat. A rating of 1 signified that *S* felt completely calm; 2, relatively calm; 3, a little uneasy; 4, quite uneasy; 5, very uneasy; and 6, extremely uneasy. An analysis of variance of these ratings (based on the average ratings for each of the 48 collections) yielded a significant sex effect ($p < .001$), threat effect ($p < .001$), and interaction between sex and threat ($p < .025$). It can be seen that the

TABLE 2
AVERAGE RATINGS OF UNEASINESS BEFORE FIRST TRIAL (EXPERIMENT I)

	Threat condition		
Sex	Low	Medium	High
Males	1.96	3.46	3.32
Females	2.22	3.55	4.38

pattern of results is generally similar to that for the number of *S*s wishing to leave the experiment. Orthogonal comparisons for the three degrees of threat show that the high and medium conditions combined are significantly higher than the low one ($p < .005$), but the high condition is not significantly higher than the medium one. The highest degree of concern about the threatened penalty is shown by the female *S*s in the high condition. Their uneasiness is significantly higher there than that of the males in the same condition ($p < .01$) and than their own uneasiness in the medium condition ($p < .01$). Apparently adding the epinephrine injection to the threat of shock produced no increment in the anxiety level of the males but did so for the females. This evidence incidentally provides a confirmation of the Schachter and Singer (1962) theory for the females but not for the males.

TABLE 3
ANALYSIS OF VARIANCE FOR PERCENTAGE ESCAPING ON FIRST TRIAL (EXPERIMENT I)

Source	Sum of squares	Degrees of freedom	Mean square	F ratio	p
Threat	44.178	2	22.089	4.710	$<.025$
Sex	60.345	1	60.345	12.861	$<.005$
Size	65.646	3	21.882	4.660	$<.025$
Threat × Sex	9.004	2	4.502	<1	
Threat × Size	22.739	6	3.790	<1	
Sex × Size	21.299	3	7.100	1.513	
Threat × Sex × Size	28.377	6	4.730	1.008	
Error	112.627	24	4.692		
Total	364.215	47			

Table 3 presents the analysis of variance for the escape data for the first trial, based upon the percentage of persons escaping in each collection during the allotted time. Percentage is used, of course, to enable comparison of the different sized collections. It can be seen that all three main effects are significant, but none of the interaction terms is. The averages for the sex and threat conditions are shown in Table 4 for easy comparison with the uneasiness ratings in Table 2. It can be seen that the percentage escaping declines as threat increases. Orthogonal comparisons among the three conditions, similar to those made for the pretrial uneasiness ratings, yield approximately similar results. High- and medium-threat conditions combined differ significantly from low threat ($p < .05$), and the difference between high and medium threat is of borderline significance ($p < .10$). Females succeed in escaping less frequently than do males. Although the sex-by-threat interaction term is not significant, the females seem more affected by the increase in threatened penalty, consistent with the pattern of results on their reported feelings of uneasiness in Table 2.

In brief, it appears that incoordination of interdependent escape increases as the threatened penalty increases. The correspondence between the pattern of escape results (in Table 4) and that of the pretrial ratings of uneasiness (in Table 2) suggests that this effect is probably mediated by the emotional reactions of anxiety or concern which are stronger in the higher threat conditions and particularly so for the female *S*s.

The average percentage escaping varies with size in the following manner: in collections of 4, 77% escape on the average; in collections of 5, 57% escape; in collections of 6, 31% escape; and in collections of 7, 49% escape. These results are generally consistent with our expectations that incoordination would increase with increasing size. It should be noted that over the range of four to six, the average *number* of persons escaping declines, even though the time permitted for escape increases. These averages, for sizes four through seven, respectively, are 3.09, 2.84, 1.83, and 3.42.

TABLE 4

AVERAGE PERCENTAGE ESCAPING ON FIRST TRIAL (EXPERIMENT I)

Sex	Threat condition		
	Low	Medium	High
Males	79	69	60
Females	59	42	11

The relatively high rate of escape in the seven-man collections is, of course, quite unexpected. That this result is not adventitious is shown by the similar result obtained by Hill from the data reported below in Experiment III. He used the same apparatus as in the present experiment and had seven real *S*s and two simulated ones. For present purposes, the latter may be disregarded because they never contributed to jams. The degree of threat, as will be shown later, was probably at about the medium threat level of the present experiment. Hill's collections yielded an average percentage of escape of 53% which is quite close to the value for the seven-man collections in the present experiment (49%).

What are the possible explanations of this unexpected effectiveness of the seven-man groups? A close consideration of the procedure used, both in this experiment and Experiment III, suggests one possibility. It will be recalled that the time allowed for the collection to escape was controlled by the flow of water from one large bottle into another. The larger the group, the more water needed in the bottle. The large volume of water necessary for the seven-man group may have been sufficiently impressive in appearance to give a degree of confidence in the adequacy of the allotted time that the smaller groups did not have. In other words, for the seven-person collections, the time may have been seen as sufficient to enable everyone to escape, this perception reducing the urgency of the need to be "first in line."

Contradicting this suggestion is the absence of any evidence of greater pretrial concern (in the ratings of uneasiness) or greater pretrial optimism (in the estimates of number who will escape) within the seven-man groups. However, it may still have been true that the reassuring effect of the large volume of water developed only after the trial began when the rate of flow became known. Experiment II was designed to eliminate this possible effect of the timing device.

EXPERIMENT II[7]

This experiment was conducted to provide further evidence on the effect of size but without the possibly contaminating effects of clues provided the *S*s as to the amount of time allotted for escape. In general, the procedure was similar to that of Experiment I except that the water-bottle timer was replaced by a tone which increased in pitch every few seconds, thus portraying the passage of time without giving any information as to when it would terminate.

Procedure

Design and subjects. Only size of the collection and sex of *S*s were varied. A single degree of threat was used throughout that was approximately equal to the medium threat condition in Experiment I. Again, collections of four, five, six, and seven persons were used, each one being composed entirely of males or females. Six collections of each sex were used for each size, yielding 48 in all and requiring 132 male and 132 female *S*s. These were undergraduates at UCLA recruited from the introductory psychology classes by a standard procedure used for most psychological experiments. This consists of circulating through the class a sign-up sheet which indicates the number and type of subjects needed and the time and place of the experiment. No indication was given of the specific nature of the experiment. Collections in the various conditions were scheduled in a random order, but departures from this order were made when fewer *S*s than planned appeared for an experimental session.

Apparatus. A shift in locale of the experiment necessitated constructing new apparatus. While it was quite different physically from that used in Experiment I, it was functionally identical. Each *S* sat before his own private console (light panel). In the top center of this was mounted a single combination push-button–indicator light unit. Below it, arrayed across the bottom of the panel, were six similar switch-

[7] This experiment was conducted by Condry in the Department of Psychology, University of California, Los Angeles.

light units.[8] The latter served simply as signal lights indicating to *S* what the other six persons (in seven-man collections) were doing and their success in escaping (if any). The top center switch-light unit served the dual purpose of showing *S* his state (whether in danger, attempting escape, or having escaped) and of providing him with his means of attempting escape (by merely pressing the push-button switch). At the beginning of a trial, all signal lights were red indicating that all *S*s were in danger. When *S* pressed his escape switch it turned yellow indicating an attempted escape. One of the lights in the bottom row, the one corresponding to this *S*, also turned yellow on each of the other consoles. After 3 seconds, if no other person interfered, the escape switch (and corresponding lights on the other consoles) turned green indicating a successful escape. If more than one *S* pressed the escape switch at the same time, both the red and yellow indicator lights appeared, indicating a jam. This combination of lights appeared on the escape switch of each person contributing to the jam and also on the indicator lights corresponding to each contributor in the bottom row of lights on all other consoles. Thus, every *S* could tell not only that there was a jam but which other *S*s were involved in it. In general, each *S* had complete information about his own and each other person's actions and fate. (However, because of a scrambled relation between seating position and position of lights on the consoles, the *S* could not tell which light corresponded to any other given participant.)

The timing and interdependency rules were applied automatically by electronic circuits. This system also deactivated a person's switch when he escaped so he could no longer interfere with the others. With collections smaller than seven, the unused switches and lights were, of course, deactivated from the outset.

The passage of time was signalled by a tone, increasing in pitch every few seconds. The *S*s were told:

> "On each trial you will all start at red, and your object is to escape (that is, to go to green) before the end of the trial is reached. When I say 'go' the trial will begin, and when I say 'stop' the trial will be over. During this time, to show you that time is passing, we'll play a tone that will sound like this. (He plays sample.) When I say 'go' the tone will start and it will slowly rise up the scale as you heard. When time is up, it will stop and I will stop and I will say 'stop.' At the end of the trial, only those people whose lights are green will have successfully escaped."

Thus, the amount of time available for the collection to escape was totally ambiguous, and there was certainly no basis for *S*s in the seven-man collections to feel any more confidence in their ability to escape than those in any other size groups. The amount of time actually allotted depended upon the size of the collection, exactly as in Experiment I.

Instructions. When the *S*s came to the experimental session they were seated in front of their individual consoles so that they could not easily see the consoles of others. (They were in separate small rooms for the most part and always at least back-to-back.) They were instructed not to look at each other nor to talk during

[8] These units are Micro Switch Series 2 combination push-button switch and indicator light devices, manufactured by Micro Switch, Freeport, Illinois. The switch components (catalog 67 number 2D64) have a two-circuit, double-break arrangement of the contacts. The indicator units (catalog 67 number 2C6) contain four lamp sockets which, with split display screens, make possible eight two-color combinations.

the session. Electrodes were taped to the first and third fingers of their nonpreferred hands. The statement of the purpose of the experiment was identical with that of Experiment I. After being told that they would be placed in a danger situation from which they were to attempt to escape, the use of the equipment and the time signal were described and demonstrated. The demonstrations were somewhat more elaborate than in Experiment I, all *S*s participating in jams and eventually being permitted to escape.

The penalty for nonescape was described as a series of painful shocks. In describing these, *E* said:

> "I don't like to have to give you these shocks, but in an experiment such as this it is necessary. To show you we will indeed give you the shock and to give you some idea of what type of shock to expect, just before we start the trial I will give you a *mild* sample shock. This will give you some idea of what is to come, but let me stress that the final shock will be much stronger."

After these instructions, the *S*s were given a mild shock (40 volts, 100 cps), and the trial was immediately begun. The purpose of this procedure was to confront the *S*s with the threat and then immediately put them into the escape situation. It was felt that by dropping them more suddenly into the action situation they would have less time to think about what to do and whether or not *E* was being truthful. (In order to achieve this, the pretrial questionnaire, which in the prior experiment had required several minutes lapse between the threat and the actual escape trial, was not used.) The pretrial shock was given because pilot data indicated *S*s were more convinced they would really be shocked after the trial if they were given a mild one before it started.

After the trial was run (and only one trial was used), the *S*s were given a posttrial questionnaire and a catharsis session. The former contained the same uneasiness rating as did Experiment I's pretrial questionnaire.

Results

For reasons given in the preceding paragraphs, the uneasiness ratings were obtained only after the trial was over, the *S*s being asked to indicate how they felt a few moments earlier, before the trial began. Nevertheless, these ratings provide our only basis for comparing the level of concern in this experiment with that in the previous one. An analysis of variance of the ratings yielded no significant size effect and only a significant sex difference ($p < .01$) with the male average being 3.79 and the female average, 4.33. Thus, as in Experiment I, the female *S*s are somewhat more concerned. The level of uneasiness indicated by these means seems most like that induced by the high-threat condition of the preceding experiment. This seems rather high in terms of a comparison of the procedures used in the two studies but is consistent with the generally high level of incoordination in the present study which, as will be seen below, also resembles that of the high-threat condition of Experiment I.

The average percentages escaping in the various conditions are given in Table 5. An analysis of variance of these results yields a significant sex difference ($p < .05$). As in Experiment I, the males have a higher rate

TABLE 5
AVERAGE PERCENTAGE ESCAPING ON FIRST TRIAL (EXPERIMENT II)

Sex	Size				Total
	Four	Five	Six	Seven	
Males	67	63	42	24	49
Females	29	37	3	19	22
Total	48	50	22	21	35

of escape than the females. The analysis of variance of the data as presented in Table 5 does not yield a significant size effect ($p < .10$). It can be seen that the size trend from the female collections is very irregular, but even with these puzzling aberrations, there seems to be a clear difference between the four- and five-man collections, on the one hand, and the six- and seven-man collections on the other. A comparison of these two halves of the size continuum yields a significant difference ($p < .05$). By combining males and females, the percentage escaping varies with size in the manner shown in bottom line of Table 5. These percentages correspond to the following average *number* of persons for the sizes four through seven, respectively: 1.92, 2.50, 1.33, and 1.50. In other words, in the larger collections, even though more time and potential escapes are available, fewer persons succeed in escaping.

When these results are compared with those of Experiment I, it is observed that the general rate of escape is lower in the present case. As noted above, this is consistent with the questionnaire data indicating that a higher level of uneasiness was achieved in this study than in the earlier one (considering all three levels of threat there).

Comparing the size trends for the two experiments, a sharp discrepancy occurs for seven-man collections where the rate of escape is markedly lower in the present data. It appears that when the clues as to available time are removed, the unexpected efficiency of the seven-man groups disappears, and the time required per escape tends to increase with group size. However, there is still a suggestion that the seven-man groups do not experience as much difficulty as the trend over the smaller groups would lead one to expect. We will discuss this later.

EXPERIMENT III[9]

Although this experiment was conducted before the two just reported, in a sense it should have followed them. In showing the important con-

[9] This experiment was conducted by Hill in the Laboratory for Research in Social Relations, University of Minnesota. It constituted the research upon which he based a dissertation submitted to the graduate faculty of the University of Minnesota in partial fulfillment of the requirements for the Ph.D.

tributions the factors of threat and size make to incoordinated collective behavior, Experiments I and II can be regarded as providing a partial validation of our laboratory setting as a simulation of natural panic situations. It then becomes appropriate to seek within the experimental setting factors that, given high threat and large size, make for more or less incoordination. Such factors might be particularly relevant in generating practical ideas about the reduction of panic behavior even in situations which, from the point of view of danger and size, are particularly prone to it.

The choice of factors for investigation here stems from the social influence analysis of the interdependent escape situation, outlined earlier. The central argument is that escape will be more coordinated when heterogeneity of the participants' initial assessments of the danger can be maintained, or at least, when they can be prevented from becoming concentrated in the state of high concern.

Susceptibility to social influence is one factor that should be related to homogenization of attitudes. An individual who is sure of the accuracy of his own assessment of the situation should be less susceptible to pressures toward uniformity than a person less sure of his own evaluation of the situation. Groups composed of individuals who are independent of social influence would be expected to maintain more successfully their original heterogeneity of assessment and action, with its concomitant advantages for escape.

The second factor investigated here is the availability of confidence responses. In the two preceding experiments, only two responses were available to *S*s, the responses of *attempting* escape and of *waiting*. Members who were confident that they would be able to escape later had no way to express this confidence to the other group members and, hence, no way to influence the unsure participants. An unsure person will use social reality as a basis for his evaluation of the danger in the situation and will look to see how others have assessed the danger in order to decide how he, himself, should react. Because the more confident persons are unable to communicate their assessments of the situation, the unsure one knows only that his fellows are either waiting passively like himself or fighting to escape. Consequently, his evaluations are likely to be influenced in the negative direction. Escape becomes embraced as the appropriate behavior and he too begins to push and fight to get out through the exit.

The provision of a special response with which the confident, unworried person can signal his assessment of the situation—a response that is easy to make, socially acceptable, and highly distinctive and recognizable —would be expected to lead to significant changes in the collection's behavior. Faced with distinctive responses made by both his more anxious and his less anxious colleagues, the uncertain individual should be able to obtain more complete information about how others regard their probable outcomes. He is less likely to be swept toward one extreme of the response continuum in a nonadaptive struggle to escape.

Procedure

Design and subjects. A 2 × 2 × 2 experimental design was used, the three factors being susceptibility to social influence (high vs. low), response availability (two responses vs. three), and sex. Data for forty collections are reported, five comprised entirely of males and five, of females in each of the four basic experimental conditions. Allocation within sex was by the method of random blocks.

The *S*s were drawn from summer session classes at the University of Minnesota. Appointments were made by telephone on the evening immediately preceding the experimental session. In the low susceptibility condition, after the *S* agreed to come at the desired time, he was told:

> "Fine! This is a group experiment. In this experiment we are able to estimate accurately how you will perform. On the basis of a few pieces of key information from your record we predict that *you* will do well. We try to make the groups up of people of differing abilities, and good people are harder to get, so that your being able to come makes my task of scheduling easier."

For *S*s scheduled for the high susceptibility condition no comparable comments were made, and in no case was any other information given about the nature of the experiment.

Apparatus. The apparatus was identical to that used in Experiment I, with two notable exceptions. First, although seven *S*s came for each experiment, there were two extra booths, one at each end of the row, which were left empty. The *S*s were unaware that these end booths were unoccupied, and the actions of *S*s supposedly in them were represented on the light panel, so the participants thought the collection consisted of nine persons. The reason for these "simulated *S*s" will be explained later. Second, the escape switch sometimes had an added function. As in the first experiment, when it was centered, at the start, a white light was on indicating "danger," and when it was moved to the right, the red "escape attempt" light went on. In the three-response condition, however, the *S* could also move the escape switch to the left and turn on a yellow light. This was used as a signal of confidence and willingness to wait. Accordingly, the column of lights for each *S* contained a yellow light in addition to the other three. Otherwise the color coding of the lights was the same as in the first experiment where white meant "danger," red meant "attempt to escape," and green meant "successful escape." The two columns of lights corresponding to the two empty booths were manipulated from a neighboring control room.

The passage of time was signaled by the flow of colored water from one bottle to another as in Experiment I. The group was given 50 seconds to escape, 42 for the seven real *S*s and 4 seconds each for the two simulated *S*s (see next section).

In the front of the room was an impressive looking "shock machine" with wires running to each booth where two electrodes were attached to the first and third fingers of the *S*s' nonpreferred hands. Although no shocks were delivered, the purpose of this machine was to make the threat of shock a real and meaningful danger.

Use of simulated subjects. A test of our ideas about the effects of a confidence response requires that there be a range of confidence within the collections. To insure that confident persons were represented it was decided to include in each group two simulated *S*s who, in the two response condition, would wait passively until all others had escaped (if this should occur) before they would attempt to escape. In the three-response condition they would express their confidence immediately and maintain this until all other group members had escaped, and only then would they proceed to escape. By this means it was possible to insure heterogeneity of expressed evaluation in the three-response condition. The simulation was carried out by leading the seven *S*s to believe that nine of them were present (it being almost impossible to tell that the two empty booths were unoccupied) and by having the assistant in the control room manipulate the two extra columns of lights, following the rules described above. In only two of the forty collections in the experiment was there a single individual who expressed suspicion about the number of *S*s present.

Instructions. The majority of the instructions were the same as those given in the two experiments already described. The *S*s were told they would be placed in a danger situation and that their task was to escape. The nature of the situation was explained, and escapes, jams, and "confidence" responses were demonstrated (the last, in the three-response condition only.) The penalty for not escaping was described as a painful but safe electric shock which would be delivered at the end of the experiment. No sample shocks were given.

High susceptibility was established by telling the *S*s that although a number of them had been present before and knew how the situation operated, the instructions were being repeated for the few who had not been there previously. This was reinstated at the end of the instructions by saying:

> Those of you who have been here before have a good idea of what is the best way to behave in this situation. For those of you who have not been here before, it would probably be best if you *watch carefully* what others are doing to help you decide the course of action which gives you the greatest chance of getting out. You can see what the other, more experienced people are doing by watching their lights up here."

Of course, none of the group members had been in the situation before, but the purpose was to make all group members feel less expert in the task than others in the group, and hence, more susceptible to the influence of the others' examples.

The *low susceptibility* manipulation, which had been introduced at the time of making the appointment, was reinstated at the beginning of the instructions as follows:

> "As I mentioned to two of you on the telephone when I called you for this experiment, we expect you two to perform well. You two should not be swayed by the way others are reacting to the situation. Make up your own minds and be independent in deciding how to deal with the situation."

At the end of the instructions they were again reminded that they should "make up their own minds." Each person, of course, was supposed to assume that he was one of the two special *S*s. This was an attempt to decrease susceptibility to social influence.

In the three-response condition, after escapes and jams had been explained and before any demonstrations had been given, the use of the extra response was explained as follows:

"This, therefore, is a situation in which consideration and politeness count. Some people have to be considerate of others and let them go first or no one will get out. By turning your switch to the left you will turn on the yellow light in the top row of your column. This signals that you have confidence that the group can escape, and you are willing to wait and let someone else escape. Thus, when you turn your switch to the left and hold it there you are indicating that you do not intend to escape immediately and that someone else should go ahead and use the exit."

Immediately before the first trial was run, the *S*s were asked to write down their estimates of how many of the nine would escape before the time limit; three trials were then run. As in the procedure outlined for Experiment I, *S*s who did not escape each time were told they would be shocked after the series of trials was over. After completion of the third trial the *S*s were given a questionnaire and then informed that no shocks would be given. A catharsis session and pledge to secrecy concluded the experiment.

Results

Level of concern. The measuring instruments used in this experiment provided no pretrial measure of uneasiness. The postexperimental questionnaire contained three questions relevant to concern. These, along with the most apt characterizations of the typical answers, are as follows: (1) "How afraid are you of electric shocks?", typically answered with "some" which fell between "very little" and "a great deal"; (2) "How sure were you during the experiment that you would be shocked if you failed to escape?", typically answered with "sure I would" which fell between "didn't know" and "very sure I would"; and (3) "How much did you worry about getting shocked today?", typically answered with "a little" which fell between "not at all" and "some." On all three questions females were significantly more concerned than males ($p < .01$ in each case). No differences were found between the response availability conditions. On the first question, high-susceptibility collections reported greater fear of shock ($p < .02$); on the second there was no susceptibility difference; and on the third, the high-susceptibility condition again indicated more worry ($p < .01$).

These results do not permit a comparison of level of concern with the other experiments, although one is inclined to believe it is somewhat lower in the present case. As in the preceding instances, females express more concern than males. (However, contrary to the other sets of data, they are *not* less efficient in escaping, as we shall see in a moment.) The variation in concern between the two susceptibility treatments would create problems for the analysis were it not for the fact that there is no relationship between a total score derived from these three questions (for each collectivity) and percentage escaping ($r = 0.08$). This makes it seem unlikely that differences in concern between the different experimental conditions could have affected the escape results. Hence, comparisons holding concern constant were not made. It should be noted that the lack of correspondence between this measure and success of escape is also inconsistent with the results from the preceding experiments.

Effect of number of available responses. Table 6 shows the average number and the average percentage (the latter based on the seven real *S*s) escaping on the first trial in each of the four experimental conditions. No sex difference in rate of escape was found, so the data for males and females have been combined in this table. A three-way analysis

TABLE 6
AVERAGE NUMBER AND PERCENTAGE ESCAPING ON FIRST TRIAL (EXPERIMENT III)

Susceptibility to influence	Number of available responses					
	Two-responses		Three-responses		Total	
	Number	Percentage	Number	Percentage	Number	Percentage
High	2.5	36	6.4	91	4.4	63
Low	2.2	31	4.3	61	3.2	46
Total	2.3	33	5.3	76	3.8	54

of variance (responses $\times$ susceptibility $\times$ sex) revealed that the only source of variance differing significantly from chance was number of responses ($F = 9.62$, $p < .005$), there being more escapes in the three-response than in the two-response condition. This is strong support for our hypothesis that providing means for the more confident participants publicly to express their feelings increases the success of the collective escape effort.

Effect of susceptibility to influence. It must first be asked whether the manipulation of susceptibility, which was a matter of differential instructions, was successful. Of three items in the postexperimental questionnaire designed to check this, two indicate partial success while the third fails to differentiate the high- and low-susceptibility groups. High-susceptibility *S*s rated themselves significantly more "influenced by what others did" ($p < .05$) and less competent and able to "perform well on this task in comparison with others in the group" ($p < .01$). Mean ratings of amount of attention paid to "what others were doing" were identical for high and lows.

From Table 6 and the analysis of variance reported above, it can be seen that the susceptibility factor does not have a simple over-all effect on success of escape. Although there is not a significant interaction term in the analysis of variance, the pattern of averages in Table 6 suggests that high susceptibility particularly promotes successful coordination when the confident members are able to express their feelings. This suggests the further possibility that the effects of susceptibility may depend upon the nature of the behavior and feelings that are predominant within the collection at the beginning of the trial. If the general tenor of feeling in the group is one of great concern, high susceptibility should make for spread of this feeling and a concentration of the *S*s in the high concern state which is hypothesized to be detrimental to successful escape. On the other hand, if the predominant initial feeling is one of confidence, high susceptibility should lead to the opposite effect.

To test this idea, we resorted to a measure of pretrial optimism for each collection based on the participants' estimates, made immediately before the trial began, of how many of the nine persons would escape. A complication arose when this estimate was found to be significantly higher (i.e., optimism was higher) in the high susceptibility condition than in the low ($p < .01$). Thus, optimism as well as susceptibility to influence was apparently manipulated by the instructions.[10] Because this seems like a situation in which peoples' expectations would tend to be self-confirming (cf. Merton, 1957, on the self-fulfilling prophecy), we might expect more escapes to occur in the more optimistic groups. Indeed, this was the case. The average optimism score within each collection was found to be correlated with the percentage of *S*s who escaped ($r = 0.34$ which, for $N = 40$, is significant at the .05 level).

These results complicated the task of examining the effects of high vs. low susceptibility while holding constant initial optimism. We proceeded as follows: the median of the average estimates of number who would escape was 5.43. The 19 collections whose average estimate was above this figure were termed *high* in optimism, and the 21 collections at or below this median were termed *low*. A cross tabulation was then made of optimism and susceptibility. The correlation between the two variables is reflected in the fact that the largest frequencies appear in the High-High and Low-Low cells. Because of this pattern, it was necessary to analyze the resulting set of data with an approximate method for the analysis of variance for unequal N's (Walker and Lev, 1953, pp. 381–382). A further complication was presented by the fact that within the high susceptibility condition, nine of the ten three-response groups were above the median in optimism, and within the

[10] On close examination of the instructions, it makes sense that the high-susceptibility *S*s were more optimistic: most *S*s except themselves were said to be experienced in the situation. In the low-susceptibility condition only *S* and one other person supposedly had the special talent; none had experience.

low-susceptibility condition, only three of the ten three-response groups were so.[11] Hence, differences due to response availability would have the effect of generating an interaction in the analysis. To eliminate this source of error, the number escaping was corrected for response availability. The mean number escaping in the three-response condition (5.3) was 3.0 greater than that in the two-response condition (2.3), so three escapes were added to each two-response entry. With this correction, Table 7 was obtained. The analysis of variance of Table 7 revealed no

TABLE 7
SUSCEPTIBILITY DIFFERENCES IN AVERAGE NUMBER ESCAPING[a] ON FIRST TRIAL, WITH OPTIMISM HELD CONSTANT (EXPERIMENT III)

Susceptibility to influence	Level of optimism		
	High	Low	Total
High	7.1 (N = 14)	3.2 (N = 6)	5.9 (N = 20)
Low	3.8 (N = 5)	4.8 (N = 15)	4.6 (N = 20)
Total	6.2 (N = 19)	4.3 (N = 21)	5.2 (N = 40)

[a] The number escaping in each collection was corrected, as described in the text, to eliminate differences due to response-availability conditions.

difference between susceptibility conditions or between levels of optimism, but there was a significant interaction between optimism and susceptibility ($p < .01$). As one would expect, when initial optimism is high the number escaping is greater in the high-susceptibility condition than in the low, but when the initial level of optimism is low, the number escaping in the low-susceptibility condition is greater than in the high.

DISCUSSION

Our results show fairly convincingly that the higher the threatened penalty for failure to escape, the greater is the incoordination (the less successful are the escape attempts). That this effect is mediated by emotional reactions to the threat is suggested by the fact that conditions in which self-ratings of uneasiness are high tend also to be those in

[11] This reflects the fact that optimism is somewhat higher in the three-response condition ($p < .10$). However, this was shown not to account for the difference between the two- and three-response conditions. The latter yielded a significantly higher rate of escape even when optimism was held constant. Thus, we can be sure that the positive effect of the three-response condition is not mediated by initially high optimism.

which escape is least successful. This holds true for the three threat conditions in Experiment I, for females vs. males in Experiments I and II, and for the general level of escape in Experiment I vs. II. The exception to this pattern occurs in Experiment III where, although the females express greater concern about getting shocked (on the posttrial questionnaire), they are no less successful in escaping than the males.

The effect of size upon incoordination is less clear. Let us disregard for the moment the puzzling results for the seven-man collections in Experiment I and treat the data as indicating a general decrease in success of escape with increasing size. Do our results contain any clues as to why this trend occurs? It will first be recalled that increasing size had no effect on pretrial ratings of uneasiness in Experiment I or upon similar posttrial ratings (made of pretrial feelings) in Experiment II. These results suggest that size did not have the preinteraction psychological effect we had thought it might, namely, that the available time would be judged less adequate for the larger collections (with resulting heightened concern). In the case of Experiment I this may reflect the special condition of the procedure in which a pretrial clue as to the available time, which varied by size, was provided by the water bottles serving as timers. However, the similar results from Experiment II where this clue was absent suggest this special condition had little effect insofar as the relation between size and *pre*trial feelings are concerned. Although the retrospective nature of the rating data in Experiment II argues for some caution in their interpretation, the lack of relation between size and rated uneasiness there and in Experiment I suggests that size has its effect *after* the trial gets under way.

Beyond suggesting the inapplicability of the pretrial effect above, our data provided only one further clue as to the basis of the size effect, one that is consistent with a simple mathematical account of the consequences of increasing size. The number of seconds required for each escape (on the average) was determined for each size by simply dividing the number of seconds allotted collections of that size by the average number of escapes (as listed in the *Results* sections for Experiments I and II). These average times were then plotted against the number of ways it is possible for jams to occur for each size. The formula for the latter is $2^n - n - 1$ for size n, which yields 11 for size four (six possible dyads, four possible trios, and one quartet), 26 for size 5, 57 for size 6, and 120 for size 7. The resulting plot is shown in Fig. 1. The data from Experiment I are represented by I's, those from Experiment II by II's, and the single data point from Experiment III by a III. It can be seen that the points from Experiment I for sizes 4, 5, and 6 fall close to a straight line. This also happens to be a line that

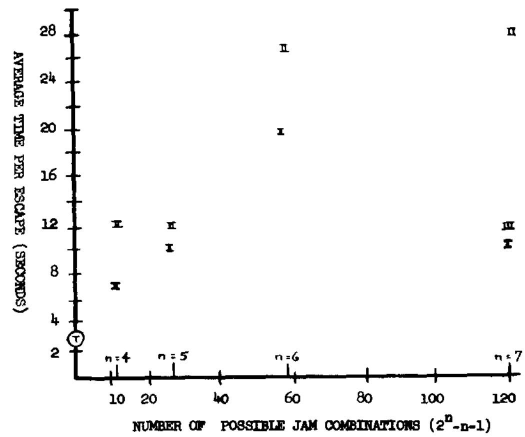

FIG. 1. Average time per escape as a function of number of possible jam combinations (all experiments).

would intersect the ordinate at a point close to what the escape time would be for a one-person "collection," shown in Fig. 1 by a "T." This "collection" would be incapable of creating jams (i.e., $2^n - n - 1 = 0$) and should be able to achieve an escape time per person of only slightly more than 3 seconds (3 seconds plus reaction time). For these sizes in Experiment I, then, time per escape seems to be a direct function of the number of possible ways jams can occur.

Unfortunately, the data points from Experiment II do not conform to this simple linear relation that seems to fit the first three points from Experiment I so well. While incoordination seems generally to increase with size in Experiment II, it is not at all clear what the probable shape of the function is. While the simple mathematical function suggested by Experiment I remains an intriguing hypothesis, further evidence is needed to determine whether (or when) a more complex function characterizes the relationship.

To return to the data with regard to seven-man collections, the difference in rates of escape for these collections between Experiments I and III on the one hand and II on the other suggests that the information provided by the water bottles in the former experiments served to facilitate coordination. However, the rate of escape for these groups even in Experiment II seems higher (time per escape being lower as in Fig. 1) than one would expect from a simple extrapolation of the results from the smaller collections.

One intriguing possible explanation is that recognition of the large number of persons who are potential candidates for use of the escape route leads some of them to give up hope of escaping. In this manner, the large group becomes functionally a smaller one. Then, as those who do attempt to escape succeed in doing so, the initially pessimistic become aroused to make attempts.

Possibly consistent with the first part of this argument is the finding that the pretrial estimates of the number that will escape, when expressed as percentages of the number of persons in the collections for which the estimates were made, tend to decline with increasing size. The data from Experiment I yield average estimates of 71% for four-man collections, 68% for five, 68% for six, and 65% for seven. Experiment III provides estimates made for nine-man collections which average 62%.

This trend might suggest that the perceived likelihood of escape declines with increasing size. On the other hand, as noted before, there is no increase in feelings of uneasiness with increasing size. The estimate of number who will escape may not reflect expectations about one's own success but merely about the fate of the collectivity. Thus, a person may estimate that few will escape but, at the same time, feel confident that he will be one of those who do. His escape attempt will then be vigorous despite his gloomy outlook for the collectivity. That this is common is suggested by data from Experiment III where an analysis was made of the correlation within each group between the individuals' optimism and their positions in the escaping queue. The evidence, significant at between the .10 and .05 level, is that the least optimistic tend to escape first.

The discussion of the last two paragraphs, in which we have proposed there may be a direct relation between a person's expectations about the likelihood of escape and his attempts to do so, suggests a distinction that perhaps should have been made earlier. This is between (1) perceived danger and (2) perceived likelihood of escape. Our discussion of the role of assessments of the situation and homogenization of attitudes was intended to be cast entirely in terms of the first factor. However, the latter factor must not be overlooked. Indeed, it was one of Mintz's (1951) major points in support of his general de-emphasis of the role of danger that, even though perceived danger is extremely high, if there is seen to be little chance of avoiding it, there will be little panic. This is the substance of the argument we have suggested (but not proved) for the seven-man groups: that some of them see so little chance of escape (at least, initially) that they effectively reduce the size of the group by their passivity in regards to escape. In general, it might be fruitful to consider these two factors as acting together multiplicatively to determine the frequency and vigor of escape attempts. If *either* perceived danger or perceived likelihood of escape are low, there will be few escape attempts. Only if both are high will the individual be oriented strongly toward attempting escape. The analysis in this paper has assumed implicitly that perceived likelihood of escape is moderately high and, in that context, has dealt with the effects of the distribution of assessments of the danger.

Experiments I and III provide data that might be considered relevant to the factor of perceived likelihood of escape: the pretrial estimate of how many would escape which we have referred to as an index of optimism. In both experiments this is positively correlated with the number who actually do escape (.40 for Experiment I and .34 for III, both significant at $p < .05$). This might be taken as evidence for the assertion above, that escape attempts are more frequent and vigorous when perceived likelihood of escape is high. However, the terms in the correlation do not correspond well to those in the theoretical statement. The relation of *escape attempts* to *actual escapes* is obviously a complex one, actual escapes being infrequent if none attempt to escape, but also, under conditions of interdependence, if too many attempt it. Similarly, as noted above, our optimism index may not reflect accurately the perceived likelihood of the individual's own escape. In short, the present data do not seem to be adequate to the task of disentangling and evaluating the role of the factors of perceived danger and perceived likelihood of escape.

The results of Experiment III suggest in several ways that it is fruitful to analyze the interdependent escape situation in terms of social influence processes and pressures toward uniformity. While the effects of our experimental manipulations of susceptibility to influence were complex, they display an eminently reasonable pattern when account is taken of initial level of concern. With high concern (low optimism), collective coordination is most adequate when persons are oriented toward disregarding each other's behavior. With low initial concern (high optimism), the opposite is true: coordination is most adequate when the participants are oriented to pay attention to each other. Another way to view this interaction is in terms of the correlation between predictions of escape and actual escapes under the two susceptibility conditions.[12] This correlation is .50 for the high-susceptibility condition (significantly different from zero with $p < .025$) and $-.30$ in the low-susceptibility

[12] The number of escapes was corrected to eliminate differential effects of the two- vs. three-response conditions, as explained earlier in the text. It must also be noted that the correlations were calculated over collections, between the *average* estimate and the number escaping, both expressed as percentages of the number in the collection.

condition (not significant). The difference between the two correlations is significant at the .02 level. Thus, the self-fulfilling prophecy effect is evidenced when susceptibility is high but not when it is low.

This result obviously has interesting but complex implications for preparing collectivities to cope successfully with interdependent escape problems. If they can be trained to have high confidence in their joint ability to escape, it is best also to teach them to be highly responsive to each other's actions. However, the latter training will be quite detrimental if they get into a situation in which their initial general level of confidence is low.

Turning to the effects of providing a means of expressing confidence during the escape period, the three-response condition was predicted to yield a higher rate of escape. This prediction was based on the assumption that a concentration of attitudes in states of high concern is detrimental to coordination of escape attempts and that availability of the confidence response would serve to maintain heterogeneity of evaluations or, at least, to prevent their converging toward the anxious states. The experimental evidence from Experiment III indicates clearly that superior coordination was achieved when the confidence response was available. To determine the basis of this effect, the data were first examined to determine whether heterogeneity of evaluations played a role in superiority. No relationship was found between within-collection *variance* on the (pretrial) optimism or (posttrial) anxiety measures and the number escaping. The lack of relationship here casts some doubt upon the interpretation that successful escape is mediated by heterogeneity. However, serious doubts must be raised about the relevance of these measures, particularly on the grounds that they may not reflect the distribution of evaluations *during* the escape period.

It is possible that the provision of three responses made the subjects more confident before the trial began and, hence, more able to escape. However, as already reported, a comparison of the response availability conditions holding initial optimism constant still yielded superiority for the three-response conditions, so this explanation seems improbable.

The possibility also exists that provision of an extra response permits better coordination of escape attempts. For example, it may be possible to resolve disagreements about position in the queue by using only the yellow ("confidence") and white ("in danger") lights without causing a jam. Thus, the participants might decide who is to be next while one of their number is in the process of escaping. The experimental records fail to support this view. There were very few groups in which the white light received any substantial use for the purpose of coordination in the three-response condition. Subjects who used the confidence light seemed then to go directly to escape (red) without waiting in the "white" state at all.

The most likely explanation of the effect of the confidence response is that the evidence of confidence it made possible reduced anxiety, increased willingness to wait, and hence reduced the amount of jamming that occurred. In other words, the public signs of confidence shifted the entire distribution of assessments toward the low anxiety end of the scale. In the low-susceptibility condition where the group members were presumably paying less attention to each other, the available public signs might be expected to make little difference, and there is some support for this. The difference between the two response conditions is smaller (although not significantly so) in the low-susceptibility condition than in the high.

The clear implication of Experiment III is that a bias in the availability of distinctive responses leads to a correspondingly biased shift in the distribution of opinion within the collectivity. That bias in available responses is not uncommon in interdependent escape settings is suggested by a close examination of Mintz's experiment and many of the real life situations it is intended to simulate. These typically provide a dramatic way for the most frightened persons to make known their evaluations of the situation (viz., the escape response) but fail to provide a single, equally distinctive response for persons who are least frightened by the threat. To be sure, there is often available a repertoire of possible responses for the latter, but there are restraints against many of these (e.g., inhibitions about making one's self conspicuous) and there is no *single* response that is highly practiced and obviously relevant. Hence, for reasons of restraint and conflict, the most confident individuals often fail to respond and their inactivity renders them indistinguishable from the middle range of persons who are most uncertain about the matter. The situation resembles, then, a group discussion in which only persons of one extreme persuasion are permitted to voice their views. The result of this seems obvious: uncertain individuals will be swayed toward the one extreme and the rate of homogenization will be much faster than if both extremes had been represented equally forcibly.

A similar explanation can probably be given for the general acceptance of extreme types of behavior in other situations in which there exists a bias in the availability of distinctive responses (e.g., in the generation of fads, fashions, and crazes). Or, to take the case of extreme political behavior, there are usually people who feel strongly that the behavior represented is undesirable. They seek for some distinctive means of dramatizing their opposition in the hope that they can swing the balance away from what they regard as inappropriate behavior. However, they often have no means of presenting their view that is as distinctive and dramatic as that of their "extremist" opponents. Here too, then, we have an example of a biased response situation in which persons at one end of an opinion continuum are not differentiated behaviorally from those with more neutral views. The present evidence would seem to indicate that lack of a distinctive and recognizable response for those of one extreme position leads to an eventual concentration of opinion at the opposite end of the scale in a portion that is not representative of the range of evaluations initially present in the group.

SUMMARY

Three experimental studies of interdependent escape, based on a laboratory simulation of one type of panic situation, have been reported. Each collection of subjects was given a limited amount of time to escape from an imminent danger, but they were able to escape only one at a time. The major findings from the experiments are:

(1) As threatened penalty for failure to escape increases, the percentage of persons who succeed in doing so declines.

(2) As size of the collection increases, the percentage escaping declines. This may also be stated as an increase in the time required per escape with increasing size. There remains considerable ambiguity about the shape of this function.

(3) If members of the collection are oriented toward taking their behavioral cues from each other, as compared with an orientation toward making their behavioral decisions independently of one another, the effect upon escape may be either a deleterious one (when at the outset there is generally little optimism about escape) or a salutary one (when the general level of initial optimism is high.)

(4) The availability of a distinctive response for the public expression of confidence greatly increases the percentage of persons who succeed in escaping.

The results are taken as indicating the fruitfulness of analyzing the interdependent escape situation in terms of present-day concepts and principles of social influence.

REFERENCES

Brown, R. W. Mass phenomena. In G. Lindzey (Ed.), *Handbook of social psychology*. Cambridge, Mass.: Addison-Wesley, 1954. Pp. 833–876.

Deutsch, M. A theory of cooperation and competition. *Human Relations,* 1949, 2, 129–152.

Festinger, L. Informal social communication. *Psychol. Rev.,* 1950, 57, 271–282.

French, J. R. P., Jr. Organized and unorganized groups under fear and frustration. *Univ. Iowa Studies Child Welfare,* 1944, **20,** 229–308.

Merton, R. K. *Social theory and social structure.* (2nd ed.). Glencoe, Ill.: Free Press, 1957.

Mintz, A. Non-adaptive group behavior, *J. abnorm. soc. Psychol.,* 1951, **46,** 150–159.

Schachter, S., and Singer, J. E. Cognitive, social, and physiological determinants of emotional state. *Psychol. Rev.,* 1962, **69,** 379–399.

Swanson, G. E., Newcomb, T. M., and Hartley, E. L. *Readings in social psychology.* (2nd ed.). New York: Henry Holt, 1952.

Walker, Helen M., and Lev, J. *Statistical inference,* New York: Henry Holt, 1953.